Mining Market Sentiment: A Comparative Analysis of Domain-Specific Language Models on Cryptocurrency News for Market Forecasting

Karl Saliba, Michel Saliba*

July 27, 2026

Abstract

The volatile nature of cryptocurrency markets presents a unique challenge for predictive forecasting, where media sentiment often drives rapid price fluctuations. This study presents a sentiment analysis pipeline evaluating four distinct natural language processing models—TextBlob, FinBERT, CryptoBERT, and Qwen2.5-7B-Instruct—on a dataset of historical cryptocurrency news spanning 2021 to 2023. We first conduct a statistical inter-comparison of the models to evaluate their baseline agreement and expose inherent, domain-specific differences in how they assess crypto-economic terminology. Subsequently, we benchmark the aggregated daily sentiment signals generated by each model against historical market data to evaluate their predictive efficacy. Our findings indicate that while Bitcoin resists isolated sentiment forecasting across all evaluated architectures, the open-weight generative model Qwen2.5-7B-Instruct attains a modest but consistent predictive edge for Ethereum and altcoin (Dogecoin) directions out-of-the-box, with an average directional accuracy of approximately 53%. Although the associated 95% confidence intervals partially overlap the 50% random-chance reference, these results suggest that compact open-weight generative models constitute a promising and computationally accessible starting point for crypto-economic sentiment analysis without the need for bespoke retraining.

1 Introduction

Financial markets are inherently volatile, but cryptocurrency markets exhibit an extreme degree of rapid and unpredictable price fluctuation [1]. Due to the lack of traditional valuation metrics, digital asset prices are heavily driven by public perception, investor behavior, and media narratives. Consequently, sentiment analysis of financial news has emerged as a powerful tool for market forecasting, with recent literature demonstrating its efficacy in extracting predictive signals from textual data [2].

Despite the proven utility of news-based market prediction, acquiring comprehensive historical datasets of cryptocurrency news remains a challenge. Researchers frequently encounter data scarcity due to URL expiration, link rot, and the ephemeral nature of digital media publications [3]. This lack of persistent, large-scale textual data makes it computationally expensive and difficult to train specialized Natural Language Processing (NLP) models from scratch. As a result, the financial technology sector often relies on pre-existing, "off-the-shelf" models. However, while some of these open-weight models were fine-tuned on traditional financial corpora, their direct applicability and accuracy when exposed to the unique, hyper-specific lexicon of the cryptocurrency domain remain uncertain.

The objective of this study is to evaluate whether existing off-the-shelf models—spanning a lexicon-based baseline, domain-specific BERT encoders, and a modern open-weight generative decoder—can be effectively utilized for cryptocurrency market prediction without requiring ground-up training. We conduct an inter-comparison of four distinct NLP models to assess their baseline agreement when scoring cryptocurrency news. Subsequently, we benchmark the sentiment signals generated by each model against actual cryptocurrency market data spanning from 2021 to 2023. This specific timeframe was selected as it represents a period of extreme macroeconomic shifts, geopolitical stress, and unprecedented boom-bust volatility within the crypto-economy [4, 5], providing a rigorous testing environment for predictive efficacy.

*michel.saliba@epfl.ch

The remainder of this paper is structured as follows. Section 2 details the methodology, outlining the selected NLP models, the dataset acquisition process, and the data analysis techniques employed. Section 3 is divided into two parts: the first presents the results of the inter-model agreement analysis, while the second provides a direct predictive comparison of the sentiment models against historical market data. Finally, Section 4 offers concluding remarks and directions for future research.

2 Methodology

This section outlines the technical framework of our study, detailing the selection of sentiment analysis models, the provenance of our historical datasets, and the statistical metrics used to evaluate model performance against market movements.

2.1 Sentiment Analysis Models

To ensure a robust comparison, we selected four models representing different architectural approaches to Natural Language Processing (NLP), ranging from a deterministic lexicon-based baseline to specialized encoder-only transformers and a large-scale generative decoder.

2.1.1 Transformer-Based Encoders: FinBERT and CryptoBERT

We utilized two domain-specific variants of the BERT architecture [6]. Both models follow the standard BERT recipe: a bidirectional transformer encoder is pre-trained on large unlabeled corpora and subsequently fine-tuned end-to-end, with a lightweight task-specific classification head producing the sentiment label.

- **FinBERT:** A model pre-trained on a large-scale financial corpus to better understand nuances in financial communication [7, 8].
- **CryptoBERT:** A RoBERTa-based model specifically fine-tuned on social media and news data from the cryptocurrency sector to capture industry-specific slang and sentiment [9, 10].

2.1.2 Dense Generative Architectures: Qwen2.5

To evaluate a generative decoder architecture, we implemented Qwen2.5-7B-Instruct, a capable open-weight model recognized for its advanced instruction-following capabilities and strong technical reasoning performance [11]. The model is a *dense* decoder-only transformer: in contrast to sparse Mixture-of-Experts (MoE) designs, which dynamically route tokens to specialized sub-networks, all of its roughly 7 billion parameters are active for every token processed, yielding a simple and predictable computational profile for batch sentiment scoring. Sentiment labels were obtained by prompting the model to classify each news item into one of the three categories used throughout this study (Positive, Neutral, or Negative).

2.1.3 Lexicon-Based Baselines: TextBlob

To establish a foundational, deterministic baseline for our sentiment analysis, we incorporated TextBlob, a widely utilized lexicon-based NLP library built upon the Natural Language Toolkit (NLTK) and Pattern [12]. Unlike transformer-based models that dynamically compute dense contextual embeddings, TextBlob operates via a predefined linguistic rule set. It analyzes textual inputs to return a tuple containing two primary metrics: polarity and subjectivity. Polarity is quantified as a continuous float score ranging from -1.0 (highly negative) to 1.0 (highly positive), while subjectivity ranges from 0.0 (strictly objective) to 1.0 (highly subjective). In this study we use only the polarity component; each article's continuous polarity score is discretized into the three categories of Equation (3) prior to aggregation. By providing a transparent, non-contextual sentiment baseline, this traditional approach allows us to quantify the performance delta—and potential contextual advantages—achieved by deploying computationally heavier, machine-learning-driven architectures.

2.2 Data Acquisition

The data used in this study is bifurcated into textual sentiment data and quantitative asset pricing data.

2.2.1 Sentiment Dataset

Historical cryptocurrency news spanning 2021 to 2023 was sourced from the "Crypto News" dataset curated by oliviervha via Kaggle [13]. This dataset provides the foundational textual inputs for our NLP pipeline, specifically utilizing timestamps, headlines, and article snippets to chronologically map market sentiment. To align this textual data with our target market assets, we filtered the corpus by its pre-assigned subject tags: articles categorized under "ethereum" were utilized to track ETH prices, articles under "blockchain" were utilized for BTC, and articles under "altcoin" were isolated to track Dogecoin (DOGE). We note that these subject tags serve as proxies for asset-specific coverage—"blockchain"-tagged news in particular spans topics well beyond Bitcoin—a simplification we revisit in Section 3.4.

2.2.2 Asset Pricing Data

Historical market data for Bitcoin (BTC), Ethereum (ETH), and Dogecoin (DOGE) were retrieved via the Yahoo Finance API using the `yfinance` Python library [14]. To align with our aggregated daily sentiment signals, we extracted the daily closing prices for each respective asset.

2.3 Data Analysis and Evaluation Metrics

Our analysis pipeline processes raw news into actionable signals using the following:

2.3.1 Market Direction Calculation

Market movement is defined as a binary classification where the direction D_t for a given day t is determined by:

$$D_t = \begin{cases} 1 & \text{if } P_{t+1} > P_t \\ 0 & \text{if } P_{t+1} \leq P_t \end{cases} \tag{1}$$

where P_t represents the closing price on day t.

2.3.2 Inter-Model Agreement

To measure the reliability of, and domain-specific divergences between, our chosen models, we utilize Cohen's Kappa coefficient (κ) [15], a chance-corrected measure of inter-rater agreement for categorical labels:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{2}$$

where p_o is the relative observed agreement between a pair of models, and p_e is the hypothetical probability of chance agreement implied by each model's marginal label distribution. Unlike raw percentage agreement, κ allows us to quantify whether two models are truly extracting the same underlying sentiment, or whether their consensus is merely incidental.

2.3.3 Sentiment Aggregation and Signal Generation

To benchmark the NLP models against binary market movements, it was necessary to map their categorical outputs (Positive, Neutral, Negative) into a quantifiable daily sentiment signal. First, each individual news article i published on day t was assigned a discrete numerical polarity score $s_{i,t}$ based on the model's classification:

$$s_{i,t} = \begin{cases} 1 & \text{if Positive} \\ 0 & \text{if Neutral} \\ -1 & \text{if Negative} \end{cases} \tag{3}$$

For any given day t containing N_t news articles, the aggregated daily sentiment score S_t was calculated as the arithmetic mean of all individual article polarities [16]:

$$S_t = \frac{1}{N_t} \sum_{i=1}^{N_t} s_{i,t} \tag{4}$$

Finally, this continuous daily sentiment score was converted into a predicted market direction $\hat{D}_t$. To align with our prior definition of actual market movement (where 1 indicates a price increase and 0 indicates a decrease or stagnation), the predictive signal was formulated as:

$$\hat{D}_t = \begin{cases} 1 & \text{if } S_t > 0 \quad \text{(Bullish Signal)} \\ 0 & \text{if } S_t \leq 0 \quad \text{(Bearish/Neutral Signal)} \end{cases} \tag{5}$$

By construction, only news published on day t is used to predict the close-to-close move from day t to day t+1. By comparing the predicted direction $\hat{D}_t$ against the actual market direction D_t, we were able to calculate the absolute predictive accuracy of each model for the selected assets.

2.3.4 Predictive Accuracy and Confidence Intervals

To evaluate the predictive efficacy of the aggregated sentiment signals against actual market movements, we utilized standard binary classification metrics. The primary evaluation metric is Classification Accuracy, which calculates the proportion of correctly predicted market directions (Up vs. Down):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{6}$$

where TP and TN represent the true positives and true negatives (correct predictions of upward and downward movements, respectively), while FP and FN represent the false positives and false negatives.

To account for sample variance across the testing period, we also calculated the Standard Error (SE) for each model's accuracy:

$$SE = \sqrt{\frac{p(1-p)}{n}} \tag{7}$$

where p is the observed accuracy proportion and n is the total number of trading days analyzed. From this, we constructed 95% confidence intervals as $p \pm 1.96 \times SE$, providing explicit error margins around each accuracy estimate.

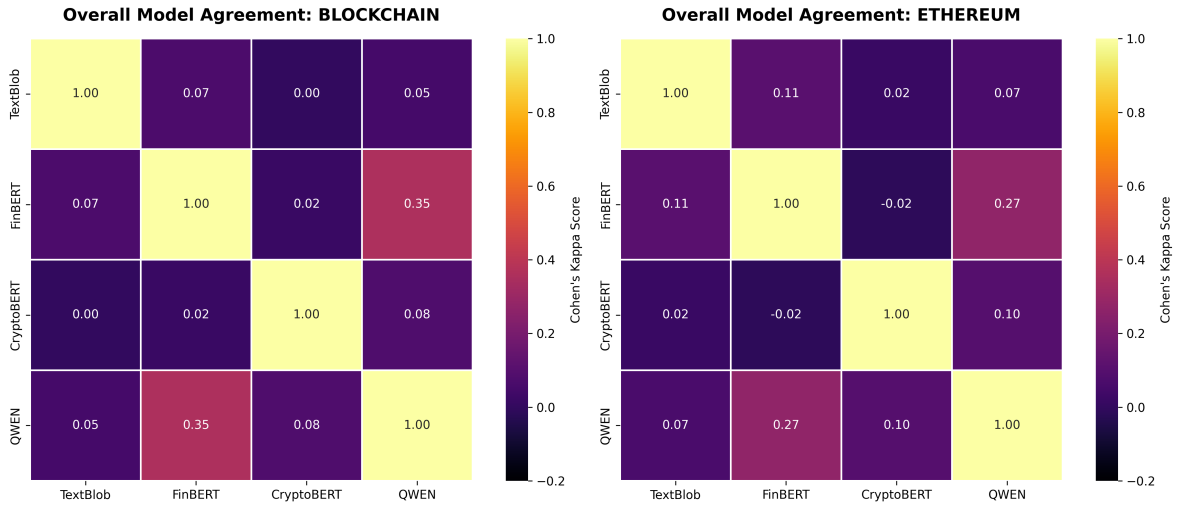
3 Results and Discussion

This section presents the empirical findings of our sentiment analysis pipeline. We first evaluate the inter-model agreement and inherent divergences across the four distinct NLP architectures. Subsequently, we benchmark their aggregated sentiment signals against actual cryptocurrency market movements to assess predictive efficacy.

3.1 Inter-Model Agreement and Bias Analysis

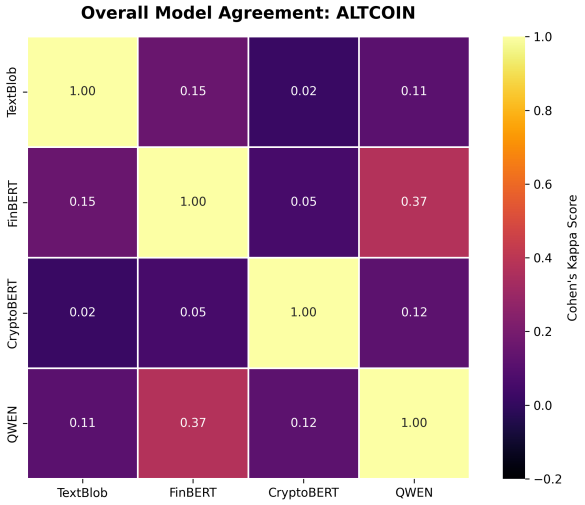
To understand how different architectures interpret the highly specialized lexicon of cryptocurrency news, we conducted a pairwise comparison using Cohen's Kappa (κ), as seen in Figure 1.

The analysis of inter-model agreement shown in Figure 1 reveals a consistent pattern across the Blockchain, Ethereum, and Altcoin sentiment datasets. The majority of model pairs exhibit near-random agreement ($\kappa \approx 0.00$), highlighting a substantial divergence in how these architectures interpret the same cryptocurrency news. A distinct exception is observed between FinBERT and Qwen2.5: across all three subjects, this pair consistently demonstrates a weak-to-moderate positive agreement ($\kappa = 0.35$, 0.27, and 0.37 on the Blockchain, Ethereum, and Altcoin subjects, respectively), suggesting that despite their structural differences, they share a more aligned reading of market context than the lexicon-based or purely crypto-native models.



(a) Blockchain (Tracking BTC)

(b) Ethereum (Tracking ETH)



(c) Altcoin (Tracking DOGE)

Figure 1: Inter-model agreement (Cohen's Kappa, κ) across three distinct cryptocurrency subjects. The heatmaps illustrate the pairwise agreement between the sentiment predictions of TextBlob, FinBERT, CryptoBERT, and Qwen2.5-7B-Instruct (labeled QWEN on the axes). To evaluate asset-specific predictive performance, the news data was filtered by subject: Blockchain news was isolated to track Bitcoin (BTC), Ethereum news for Ether (ETH), and Altcoin news for Dogecoin (DOGE). Diagonal values represent perfect self-agreement ($\kappa = 1.00$). Off-diagonal values reveal substantial divergences in model behavior; notably, TextBlob and CryptoBERT exhibit near-random agreement ($\kappa \approx 0.00$) with all other models, while FinBERT and Qwen2.5 demonstrate a weak-to-moderate positive agreement.

3.2 Market Predictive Efficacy

The ultimate utility of financial sentiment analysis lies in its predictive power. We evaluated the daily aggregated sentiment signal of each model against the next-day binary price direction of Bitcoin (BTC), Ethereum (ETH), and Dogecoin (DOGE); the resulting accuracies are summarized in Figure 2.

For BTC, no model meaningfully exceeded chance (TextBlob: 51.84%; FinBERT: 51.13%; CryptoBERT: 48.02%; Qwen2.5: 49.72%), and every 95% confidence interval straddled the 50% line. For ETH, Qwen2.5 achieved the highest accuracy at 53.44%, followed by FinBERT at 52.21%, with CryptoBERT at 50.08% and TextBlob below chance at 49.16%. For DOGE, Qwen2.5 (53.28%) and TextBlob (52.88%) led the field, while FinBERT (50.60%) and CryptoBERT (50.47%) remained close to chance.

3.3 Discussion

In evaluating the predictive efficacy of our sentiment pipelines, distinct performance tiers and asset-specific limitations emerged. On average, the domain-specific CryptoBERT exhibited the lowest predictive accuracy, underperforming even the traditional, lexicon-based TextBlob baseline. One plausible contributor is a train–inference domain mismatch: CryptoBERT was fine-tuned predominantly on short, informal social-media posts, whereas our corpus consists of professionally written news headlines and snippets. While FinBERT generated a mean predictive accuracy marginally above the 50% random baseline, its 95% confidence interval intersected with chance, rendering the performance edge statistically insignificant.

Notably, forecasting the Bitcoin (BTC) market proved universally challenging across all evaluated architectures. This uniform underperformance suggests that as a highly capitalized, macro-integrated asset, Bitcoin's price action is driven by a complex, multi-faceted blend of global economic factors that isolated news sentiment cannot independently capture.

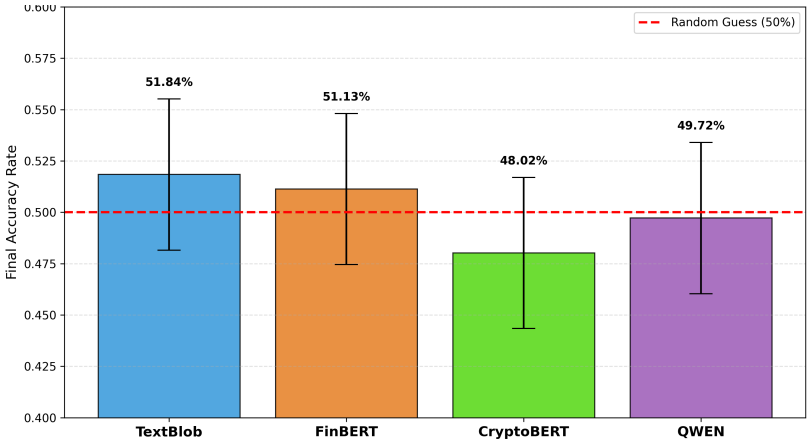
Conversely, the dense generative model Qwen2.5-7B-Instruct demonstrated the strongest out-of-the-box performance on the Ethereum (ETH) and altcoin (Dogecoin) datasets, with an average accuracy of approximately 53%. Its point estimates sit above the 50% random-guess threshold on both assets, although the lower bounds of the corresponding 95% confidence intervals remain close to—and partially below—chance. These findings indicate that, while apex market-cap assets appear to require multidimensional forecasting inputs, modern open-weight generative models such as Qwen2.5 can extract a modest predictive sentiment signal for secondary and alternative cryptocurrencies without bespoke retraining. We regard this edge as promising rather than conclusive, pending the stronger baselines and formal significance testing discussed below.

3.4 Limitations

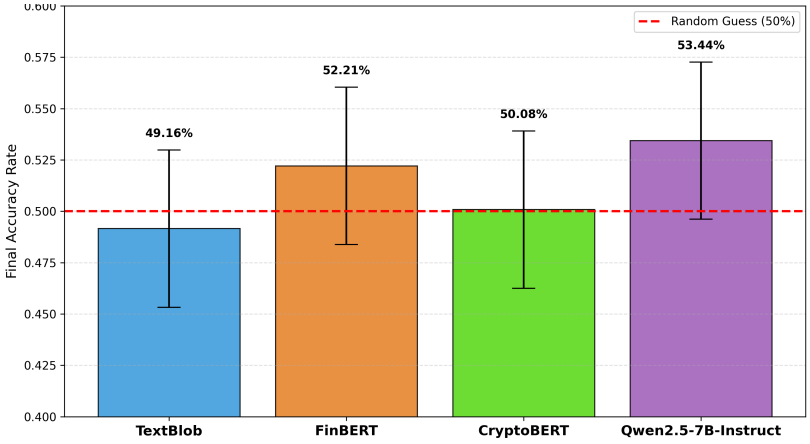
Several limitations qualify these results. First, the 50% reference line is a floor rather than a competitive baseline: because 2021–2023 contained pronounced directional regimes (a bull market, a prolonged drawdown, and a recovery), the empirical balance of up- versus down-days for each asset need not equal 50/50, and a meaningful edge must be demonstrated against the per-asset majority-class rate and against a simple persistence rule (predicting that today's direction repeats tomorrow), not only against a coin flip. Second, with four models evaluated on three assets, twelve accuracy estimates are implicitly compared against chance, inflating the probability that some exceed 50% by luck alone; formal tests with multiple-comparison corrections are required before claiming significance. Third, the subject-tag-to-asset mapping (e.g., "blockchain"-tagged news as the Bitcoin signal) is a coarse proxy for asset-specific coverage. Fourth, directional accuracy does not imply profitability: no transaction costs, slippage, or position sizing are modeled here, and an economic backtest is left to future work. Finally, all results derive from a single news source and from fixed prompting and thresholding choices whose sensitivity we have not yet quantified.

4 Conclusion

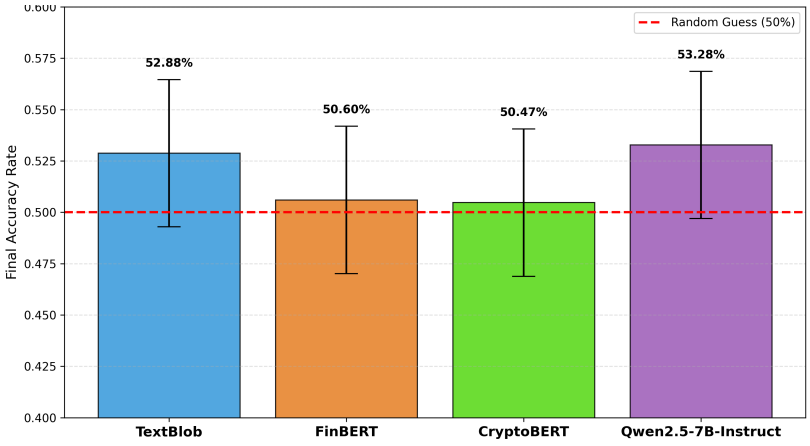
This study addressed the significant challenge of cryptocurrency market forecasting in an environment characterized by extreme volatility and historical data scarcity. Because the ephemeral nature of digital financial media makes training bespoke NLP models from scratch highly restrictive and computationally expensive, our primary objective was to determine whether existing, off-the-shelf architectures



(a) Blockchain Sentiment vs. BTC Prices



(b) Ethereum Sentiment vs. ETH Prices



(c) Altcoin Sentiment vs. DOGE Prices

Figure 2: Directional predictive accuracy of the four sentiment models across three distinct cryptocurrency markets. Bar heights represent the final aggregate accuracy of each model predicting the $T+1$ daily market direction, evaluated against a 50% random-chance reference (red dashed line). Error bars denote the 95% confidence intervals ($\pm 1.96 \times SE$) for each prediction set.

could be effectively deployed out-of-the-box. By evaluating a diverse pipeline of domain-specific encoders, a generative decoder, and a lexicon-based baseline on a 2021–2023 news dataset, we tested their capacity to extract actionable market signals without requiring ground-up retraining.

Our inter-model comparison revealed distinct behavioral differences among the architectures when exposed to the hyper-specific crypto lexicon. Notably, the Cohen's Kappa analysis demonstrated near-random agreement ($\kappa \approx 0.00$) among the majority of model pairs, highlighting a substantial divergence in how legacy and purely crypto-native models interpret the same text. However, a distinct exception emerged between FinBERT and Qwen2.5. Across all three subjects, this pair consistently exhibited a weak-to-moderate positive agreement (reaching $\kappa = 0.37$ on the Altcoin subject), suggesting that despite their structural differences, both models share a more aligned understanding of underlying financial risk and market context.

When these aggregated sentiment scores were benchmarked against actual price action, predictive efficacy varied by asset class. Predicting Bitcoin's direction proved resistant to isolated sentiment forecasting across all architectures, whereas Qwen2.5-7B-Instruct emerged as the most promising out-of-the-box forecaster for the Ethereum and altcoin markets. While the lower tails of its 95% confidence intervals dipped below the coin-flip threshold, its point estimates remained above the 50% reference on both assets. Conversely, the crypto-native CryptoBERT demonstrated the weakest overall predictive performance, consistent with its near-zero agreement with every other model.

Ultimately, this study provides preliminary evidence that compact open-weight generative models such as Qwen2.5-7B-Instruct offer a viable, computationally accessible entry point for crypto-economic sentiment analysis in alternative-coin markets, mitigating the bottleneck of historical data scarcity and bespoke retraining. Future work should (i) benchmark the sentiment signals against majority-class and persistence baselines with formal significance testing, (ii) broaden the model set to additional generative families and a proprietary frontier model as an upper bound, (iii) evaluate economic significance through cost-aware backtesting, and (iv) explore the real-time streaming integration of these models into live algorithmic trading pipelines.

References

[1] İlayda İSABETLİ FİDAN and Tuğba GÜZ. The characteristics of cryptocurrency market volatility: Empirical study for five cryptocurrency. *Alphanumeric Journal*, 10(2):69–84, December 2022. ISSN 2148-2225. doi: 10.17093/alphanumeric.941529. URL http://dx.doi.org/10.17093/alphanumeric.941529.

[2] Zuzana Janková. Critical review of text mining and sentiment analysis for stock market prediction. *Journal of Business Economics and Management*, 24(1):177–198, April 2023. ISSN 2029-4433. doi: 10.3846/jbem.2023.18805. URL http://dx.doi.org/10.3846/jbem.2023.18805.

[3] Kelly Ann Coulter. The impact of news media on bitcoin prices: modelling data driven discourses in the crypto-economy with natural language processing. *Royal Society Open Science*, 9(4), April 2022. ISSN 2054-5703. doi: 10.1098/rsos.220276. URL http://dx.doi.org/10.1098/rsos.220276.

[4] Raphael Auer, Giulio Cornelli, Sebastian Doerr, Jon Frost, Leonardo Gambacorta, Jon Frost, and Leonardo Gambacorta. Crypto trading and bitcoin prices: Evidence from a new database of retail adoption. *SSRN Electronic Journal*, 2023. ISSN 1556-5068. doi: 10.2139/ssrn.4357559. URL http://dx.doi.org/10.2139/ssrn.4357559.

[5] Mengyu Wang, Shay B Cohen, and Tiejun Ma. Modeling news interactions and influence for financial market prediction. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, page 3302–3314. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.findings-emnlp.189. URL http://dx.doi.org/10.18653/v1/2024.findings-emnlp.189.

[6] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding, 2018. URL https://arxiv.org/abs/1810.04805.

[7] Dogu Araci. Finbert: Financial sentiment analysis with pre-trained language models, 2019. URL https://arxiv.org/abs/1908.10063.

[8] Prosus AI. Finbert: Financial sentiment analysis with pre-trained language models. https://huggingface.co/ProsusAI/finbert, 2019. Accessed via Hugging Face model hub.

[9] Mikolaj Kulakowski and Flavius Frasincar. Sentiment classification of cryptocurrency-related social media posts. *IEEE Intelligent Systems*, 38(4):5–9, 2023. ISSN 1941-1294. doi: 10.1109/mis.2023.3283170. URL http://dx.doi.org/10.1109/MIS.2023.3283170.

[10] ElKulako. Cryptobert. https://huggingface.co/ElKulako/cryptobert, 2024. Accessed via Hugging Face model hub.

[11] Qwen Team. Qwen2.5-7b-instruct. https://huggingface.co/Qwen/Qwen2.5-7B-Instruct, 2024. Accessed via Hugging Face model hub.

[12] Steven Loria. Textblob: Simplified text processing. https://textblob.readthedocs.io/en/dev/, 2018. Accessed for rule-based sentiment baseline.

[13] olivierva. Crypto news dataset. https://www.kaggle.com/datasets/oliviervha/crypto-news, 2023.

[14] Ran Aroussi. yfinance: Yahoo! finance market data downloader. https://github.com/ranaroussi/yfinance, 2020.

[15] Jacob Cohen. A coefficient of agreement for nominal scales. *Educational and psychological measurement*, 20(1):37–46, 1960.

[16] Thanos Konstantinidis, Giorgos Iacovides, Mingxue Xu, Tony G. Constantinides, and Danilo Mandic. Finllama: Financial sentiment classification for algorithmic trading applications, 2024. URL https://arxiv.org/abs/2403.12285.